

Assignment 2 A

1. Configure the git in vs code

```
git config --global user.name "Your Full Name"
```

```
git config --global user.email "youremail@example.com"
```

2. git init

3. git add . // add all the file in the currecnt directory

4. git commit -m "Initial commit"

5. git remote add origin https://github.com/yourusername/my-project.git

6. git branch // check branch if it is master then:

change it to main : git branch -m master main

7. git branch -M main

8. git push -u origin main

Assignmnet 3 A

1. check the node and npm is installed in pc:

a. node -v

b. npm -v

2. npm init -y // initilize the node in project

3. npm install express // used for api creation

4. create a public folder in project and add index,html file:

```
<!DOCTYPE html>
```

```
<html lang="en">
```

```
<head>
```

```
<meta charset="UTF-8">
```

```
<title>My Static Website</title>
```

```
</head>
```

```
<body>
```

```
<h1>Welcome to My Static Website</h1>

<p>This page is served using Node.js and Express.</p>

</body>

</html>
```

5.create the server.js file in project folder

```
const express = require('express');
const app = express();
const PORT = 3000;

// Serve static files from 'public' folder
app.use(express.static('public'));

app.listen(PORT, () => {
  console.log(` Server is running at http://localhost:${PORT} `);
});
```

run the server : node server.js

Assignmnet 3 B

Steps to create backend

1. create a backend folder

2. cd backend

3. npm init -y // initilize node project

4. install dependences :

a. express – Web framework for building APIs

npm install express

b. mongoose – Connect and interact with MongoDB

npm install mongoose

c. cors – Enable cross-origin requests

npm install cors

d. body-parser – Parse incoming request bodies (e.g., JSON)

npm install body-parser

e. bcryptjs – Hash and compare passwords securely

npm install bcryptjs

f. jsonwebtoken – Create and verify JWT tokens for auth

npm install jsonwebtoken

#g. dotenv - link .env file to server

npm install dotenv

5.create

backend/

├── models/

| └── User.js

├── routes/

| ├── auth.js

| └── profile.js

├── .env

├── server.js

└── package.json

6. .env

PORT=5000

MONGO_URI=mongodb://127.0.0.1:27017/userAuthDB

JWT_SECRET=yourSecretKey

7.models/User.js

```
const mongoose = require('mongoose');
```

```
const userSchema = new mongoose.Schema({  
  name: String,  
  email: { type: String, unique: true },  
  password: String  
});
```

```
module.exports = mongoose.model('User', userSchema);
```

8.server.js

```
const express = require('express');
```

```
const mongoose = require('mongoose');
```

```
const cors = require('cors');
```

```
require('dotenv').config();
```

```
const authRoutes = require('./routes/auth');
```

```
const profileRoutes = require('./routes/profile');
```

```
const app = express();
```

```
app.use(cors());
```

```
app.use(express.json());
```

```

// MongoDB connection
mongoose.connect(process.env.MONGO_URI, {
  useNewUrlParser: true,
  useUnifiedTopology: true,
}).then(() => console.log('MongoDB connected'))
  .catch(err => console.error('MongoDB error:', err));

// Routes
app.use('/api/auth', authRoutes);
app.use('/api/profile', profileRoutes);

// Start server
const PORT = process.env.PORT || 5000;
app.listen(PORT, () => console.log(` Server running on port ${PORT} `));

9.routes/auth.js
const express = require('express');
const bcrypt = require('bcryptjs');
const jwt = require('jsonwebtoken');
const User = require('../models/User');

const router = express.Router();
const JWT_SECRET = process.env.JWT_SECRET;

// Register
router.post('/register', async (req, res) => {
  const { name, email, password } = req.body;
  const userExists = await User.findOne({ email });
  if (userExists) return res.status(400).json({ message: "User already exists" });

```

```

    const hashedPassword = await bcrypt.hash(password, 10);
    const user = new User({ name, email, password: hashedPassword });
    await user.save();
    res.json({ message: "User registered" });
  });

// Login
router.post('/login', async (req, res) => {
  const { email, password } = req.body;
  const user = await User.findOne({ email });
  if (!user) return res.status(400).json({ message: "User not found" });

  const isMatch = await bcrypt.compare(password, user.password);
  if (!isMatch) return res.status(400).json({ message: "Invalid credentials" });

  const token = jwt.sign({ id: user._id }, JWT_SECRET, { expiresIn: '1h' });
  res.json({ token });
});

```

```

module.exports = router;

```

10.routes/profile.js

```

const express = require('express');
const jwt = require('jsonwebtoken');
const User = require('../models/User');

const router = express.Router();
const JWT_SECRET = process.env.JWT_SECRET;

```

```
// Profile route (protected)
router.get('/', async (req, res) => {
  const token = req.headers.authorization?.split(' ')[1];
  if (!token) return res.status(401).json({ message: "Unauthorized" });

  try {
    const decoded = jwt.verify(token, JWT_SECRET);
    const user = await User.findById(decoded.id).select("-password");
    res.json(user);
  } catch (err) {
    res.status(401).json({ message: "Invalid token" });
  }
});

module.exports = router;
```

11. start backend : node server.js

Create to frontend

1.npx create-react-app user-auth-app

2.cd user-auth-app

3.install dependences

a.npm install axios

b.npm install react-router-dom

4.src/

```
|
|
|└─ components/
|  |└─ Register.js
|  |└─ Login.js
|  └─ Profile.js
|
└─ App.js
```

5.app.js

```
import React from 'react';
import { BrowserRouter as Router, Routes, Route, Link } from 'react-router-dom';
import Register from './components/Register';
import Login from './components/Login';
import Profile from './components/Profile';
```

```
function App() {
  return (
    <Router>
      <div>
        <nav>
          <Link to="/register">Register</Link> |
          <Link to="/login">Login</Link> |
          <Link to="/profile">Profile</Link>
        </nav>
        <Routes>
          <Route path="/register" element={<Register />} />
          <Route path="/login" element={<Login />} />
          <Route path="/profile" element={<Profile />} />
        </Routes>
      </div>
    </Router>
  );
}
```



```

    </Routes>
  </div>
</Router>
);
}

```

```
export default App;
```

6.Register.js

```

import React, { useState } from 'react';
import axios from 'axios';

```

```

function Register() {
  const [user, setUser] = useState({ name: '', email: '', password: '' });

```

```

  const handleSubmit = (e) => {
    e.preventDefault();
    axios.post('http://localhost:5000/register', user)
      .then(res => alert("Registered Successfully!"))
      .catch(err => console.log(err));
  };

```

```

  return (
    <form onSubmit={handleSubmit}>
      <h2>Register</h2>
      <input type="text" placeholder="Name" onChange={e => setUser({ ...user, name:
e.target.value }}} required />
      <input type="email" placeholder="Email" onChange={e => setUser({ ...user, email:
e.target.value }}} required />

```

```

    <input type="password" placeholder="Password" onChange={e => setUser({ ...user,
password: e.target.value })} required />

    <button type="submit">Register</button>

  </form>

);
}

```

```
export default Register;
```

7.Login.js

```
import React, { useState } from 'react';
```

```
import axios from 'axios';
```

```
function Login() {
```

```
  const [user, setUser] = useState({ email: '', password: '' });
```

```
  const handleSubmit = (e) => {
```

```
    e.preventDefault();
```

```
    axios.post('http://localhost:5000/login', user)
```

```
      .then(res => {
```

```
        alert("Login Success!");
```

```
        localStorage.setItem('token', res.data.token);
```

```
      })
```

```
      .catch(err => alert("Invalid credentials"));
```

```
  };
```

```
  return (
```

```
    <form onSubmit={handleSubmit}>
```

```
      <h2>Login</h2>
```

```
    <input type="email" placeholder="Email" onChange={e => setUser({ ...user, email:
e.target.value })} required />
```

```
    <input type="password" placeholder="Password" onChange={e => setUser({ ...user,
password: e.target.value })} required />
```

```
    <button type="submit">Login</button>
```

```
  </form>
```

```
);
```

```
}
```

```
export default Login;
```

8.Profile.js

```
import React, { useEffect, useState } from 'react';
```

```
import axios from 'axios';
```

```
function Profile() {
```

```
  const [profile, setProfile] = useState({});
```

```
  useEffect(() => {
```

```
    const token = localStorage.getItem('token');
```

```
    axios.get('http://localhost:5000/profile', {
```

```
      headers: { Authorization: `Bearer ${token}` }
```

```
    })
```

```
    .then(res => setProfile(res.data))
```

```
    .catch(err => console.log(err));
```

```
  }, []);
```

```
  return (
```

```
    <div>
```

```
      <h2>Profile</h2>
```

```

{profile.name ? (
  <div>
    <p>Name: {profile.name}</p>
    <p>Email: {profile.email}</p>
  </div>
) : (
  <p>Loading or unauthorized</p>
)}
</div>
);
}

```

export default Profile;

9. start server : npm start

Assignmnet 4 A

1. create a index.html file and add cnd link of jQuery Moblie in head

```

<!-- jQuery Mobile CSS -->
<link rel="stylesheet" href="https://code.jquery.com/mobile/1.4.5/jquery.mobile-1.4.5.min.css" />
<!-- jQuery (necessary for jQuery Mobile) -->
<script src="https://code.jquery.com/jquery-3.6.0.min.js"></script>
<!-- jQuery Mobile JavaScript -->
<script src="https://code.jquery.com/mobile/1.4.5/jquery.mobile-1.4.5.min.js"></script>

```

2. add the other code in index.html file

```
<head>

  <link rel="stylesheet" href="style.css">

</head>

<body>

  <!-- Home Page -->

  <div data-role="page" id="home">

    <div data-role="header">

      <h1>Welcome to My Mobile Website</h1>

    </div>

    <div data-role="content">

      <p>This is a simple mobile website built using jQuery Mobile.</p>

      <a href="#about" class="ui-btn ui-corner-all">Go to About</a>

    </div>

  </div>

  <!-- About Page -->

  <div data-role="page" id="about">

    <div data-role="header">

      <h1>About Us</h1>

    </div>

    <div data-role="content">

      <p>This is the About page. You can add more information about your site here.</p>

      <a href="#home" class="ui-btn ui-corner-all">Back to Home</a>

    </div>

  </div>

</body>
```

</html>

3. explanations :

a.data-role="page" represents a full-screen mobile page

b.<div data-role="header">: Defines the header section of your page

c.<div data-role="content">: This is the main content

4. if want to add more features of jqury check : <https://demos.jquerymobile.com/1.4.5/>